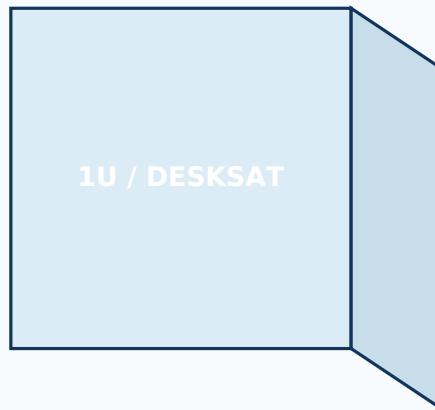


CUBESAT HEALTH MONITORING SYSTEM

A DIY Engineering Testbed for Telemetry, Fault Detection and Ground-Station GUI



Engineering Design and Research Report

Focus: EPS + TCS + OBC + ADCS + COMMS + Payload + Ground Station

Prototype level: high-quality college engineering / Hardware-in-the-Loop

Abstract

This report proposes a high-quality, laboratory-scale CubeSat health monitoring system (HMS) suitable for a college engineering capstone and for development toward a research-paper-quality experimental platform. The project emulates spacecraft housekeeping telemetry using an embedded on-board computer, an electrical power subsystem, thermal sensors, inertial sensing, a communications link, a mission payload, and a ground-station graphical user interface (GUI). The architecture emphasizes measurable engineering performance rather than a hobby demonstration: calibrated sensing, deterministic telemetry generation, packet integrity checking, rule-based fault classification, data logging, alarm handling, and hardware-in-the-loop (HIL) experimentation are treated as first-class requirements. The report recommends industrial-quality COTS components for the prototype, while explicitly distinguishing them from flight-qualified space hardware. It also provides a validation plan, suggested telemetry limits, research questions, and a proposed paper structure.

Keywords: CubeSat, housekeeping telemetry, health monitoring system, electrical power subsystem, thermal monitoring, ADCS, CCSDS, embedded systems, RTOS, ground station, anomaly detection, hardware-in-the-loop.

1. Project Definition

The proposed system is a 1U-class DeskSat laboratory platform whose purpose is to answer a practical engineering question: Can a low-cost but high-quality terrestrial testbed reproduce the essential telemetry, health assessment, and operator-interface behaviors expected from a small spacecraft? The project does not claim to be a flight-ready CubeSat. Instead, it creates a traceable test environment in which sensing, embedded processing, communications, visualization, and fault handling can be quantitatively evaluated.

The design is intentionally subsystem-oriented. EPS reports electrical power and battery behavior; TCS monitors internal and external temperatures; OBC executes the real-time telemetry and health logic; ADCS sensors supply inertial measurements; COMMS transports telemetry; the payload creates mission-like data; and the ground station decodes, stores, trends, and presents the information. This mirrors the systems-engineering thinking used for small-spacecraft development while remaining achievable in a university laboratory.

CubeSats use standardized unit geometry and emerged as an educational and technology-demonstration platform. NASA describes the 1U nominal unit as 10 x 10 x 10 cm and notes that larger configurations such as 1.5U, 2U, 3U, 6U and 12U are common. NASA also notes that the physical CubeSat structure can extend beyond the nominal unit envelope; current smallsat guidance, for example, lists a typical 1U primary structure around 100 x 100 x 113.5 mm. [\[cite\]turn453057search7\[turn453057search10\]](#)

2. System Objectives

	Objective	Acceptance Criteria
O1	Acquire health telemetry	Stable sampling from EPS, TCS, ADCS and OBC sensors with timestamps.
O2	Assess health automatically	Nominal / Warning / Critical state based on limits, persistence and sensor validity.
O3	Transmit telemetry reliably	Framed packets with sequence number, timestamp, payload and CRC; packet-loss measured experimentally.
O4	Provide operator awareness	Ground GUI with current state, trends, alarms, acknowledgements and event history.
O5	Support bidirectional control	Safe telecommand path for selected non-destructive commands.
O6	Enable research	Repeatable HIL experiments producing quantitative latency, integrity, stability and fault-detection results.

A research contribution can then be framed around a specific evaluation problem, such as adaptive health scoring, multi-parameter anomaly detection, or the effect of packet loss and sensor uncertainty on ground-station health decisions.

3. System Architecture

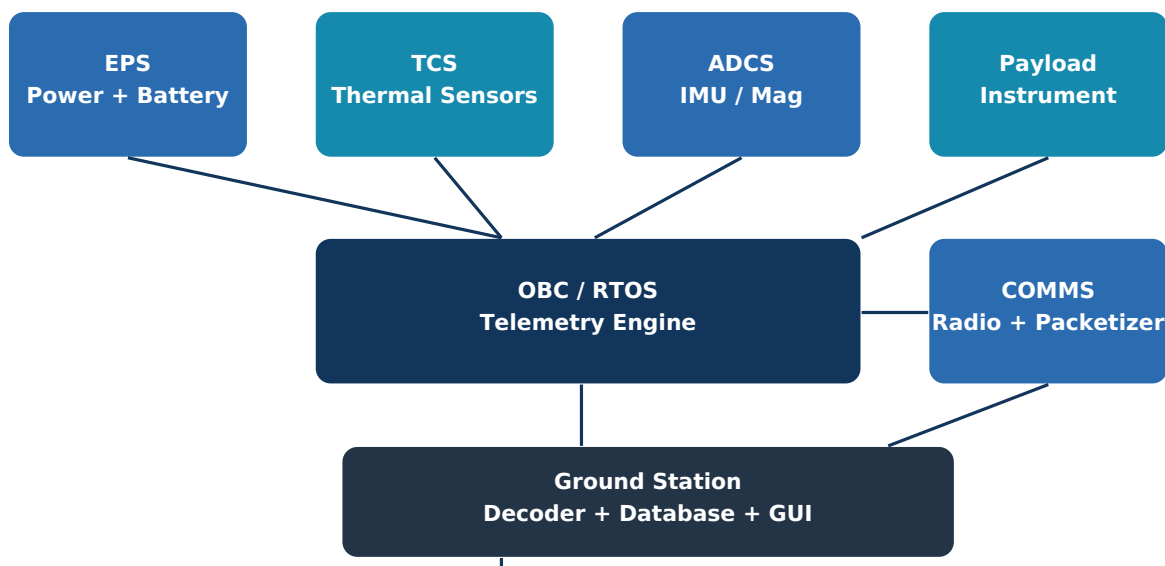


Figure 1. Proposed end-to-end architecture for the CubeSat health monitoring demonstrator. The OBC is the central telemetry and health-assessment node; the ground station remains responsible for operator-level visualization, archival logging and analysis.

The architecture follows a clean separation of concerns. Sensor drivers collect raw values. A calibration layer converts them into engineering units and tracks sensor validity. A health engine evaluates thresholds and derived indicators. The telemetry manager creates fixed-format packets. A communications abstraction supports USB serial, UART or an RF module without changing higher-level application code. The ground station mirrors this separation through transport, decoding, validation, storage, analytics and GUI layers.

A key research-quality decision is to avoid making the GUI itself the health engine. Critical limits should be represented in the satellite-side data model so that the same logic can be tested independently of the display. The GUI can then display the satellite decision and, optionally, a second ground-side diagnostic calculation.

4. Health Parameters and Recommended Monitoring Set

Subsystem	Parameter	Type	Health Interpretation
EPS	Battery voltage	Analog	Undervoltage, overvoltage, rate-of-change, plausibility
EPS	Battery current	Analog	Charge/discharge magnitude and unexpected load
EPS	Bus voltage (3.3/5 V)	Analog	Rail regulation and brownout risk
EPS	Bus current / power	Analog	Per-rail loading and power budget
EPS	Battery temperature	Analog	Thermal safety and charge/discharge envelope
EPS	Solar input voltage/current	Analog	Generation estimate and charging opportunity
OBC	CPU utilization	Derived	Scheduling margin and overload indication
OBC	RAM / storage utilization	Derived	Resource exhaustion risk

Subsystem	Parameter	Type	Health Interpretation
OBC	Uptime / reset count	Event	Unexpected reboot or watchdog activity
OBC	Task heartbeat counters	Event	Missing task / scheduler fault
TCS	Internal temperature	Analog	Thermal state of electronics
TCS	Panel / external temperature	Analog	External thermal boundary behavior
ADCS	Gyro x/y/z	Analog	Angular motion and sensor health
ADCS	Accel x/y/z	Analog	Motion, vibration and attitude cues
ADCS	Magnetometer x/y/z	Analog	Magnetic field vector and sensor plausibility
COMMS	RSSI / link quality	Analog	Receiver margin and connectivity
COMMS	Packet sequence gaps	Derived	Packet loss and link reliability
COMMS	CRC failures	Derived	Data corruption / noise indicator
COMMS	Transmit queue depth	Derived	Backpressure / data backlog
Payload	Power state	Discrete	Instrument availability
Payload	Data queue depth	Derived	Mission-data storage / downlink pressure
System	Fault flags	Discrete	Aggregated health state by subsystem

5. Health State Model

A useful GUI should not display raw values without context. Each telemetry parameter should be mapped to a state using a rule that reflects the engineering meaning of the quantity. A three-level model is adequate for the first implementation: Nominal, Warning and Critical. A fourth state, Invalid, should be reserved for missing, stale or impossible sensor data.

State	Example rule	Operative meaning	System action
NOMINAL	Inside validated operating band	Healthy	Continue normal telemetry
WARNING	Near limit, persistent drift, or elevated uncertainty	Investigate	Raise banner; increase logging; optional mitigation
CRITICAL	Hard limit exceeded or protective event	Unsafe / degraded	Immediate alarm; inhibit selected loads or enter safe mode
INVALID	No update, CRC fail, out-of-range raw value	Unknown	Do not trust measurement; flag stale/invalid

For a research-grade implementation, thresholding should include hysteresis and persistence. For example, the battery could enter Warning when voltage falls below a configurable threshold for ten consecutive samples and return to Nominal only after it rises above a higher recovery threshold for a specified period. This avoids alarm chatter. The same pattern can be applied to temperatures, bus voltages and communication errors.

6. Derived Health Metrics

Metric	Calculation	Key Information
State of Charge (SoC)	Battery-specific estimate based on voltage/current and, ideally, coulomb counting	Separates energy availability from instantaneous voltage.
Power balance	$P_{\text{generated}} - P_{\text{consumed}}$	Identifies whether the system is energetically stable.
Thermal margin	$T_{\text{limit}} - T_{\text{measured}}$	Shows distance to a thermal boundary.

Metric	Context	How it matters
Packet loss rate	Missing sequence numbers / expected packets	Quantifies link quality independent of RSSI alone.
Telemetry latency	Ground arrival time - source timestamp	Measures end-to-end operator awareness.
Sensor validity score	Range, rate-of-change and communication checks	Prevents corrupt telemetry from becoming false alarms.
Subsystem health score	Weighted / rule-based aggregation	Supports a single mission-level status indicator while preserving detail.

7. Recommended Hardware for a High-Quality DIY Engineering Prototype

The hardware should be COTS and well-documented, but the report should never label catalog parts as “space-qualified” unless the specific procurement and qualification basis supports that claim. The recommended stack is therefore described as industrial-quality prototype hardware, not flight hardware. NASA’s CubeSat mission-success guidance emphasizes that small-spacecraft projects are strongly constrained by size, compactness, cost and schedule, which makes this separation between prototype and flight qualification especially important. [cite]turn453057search4[

Subsystem	Recommended device / value	Relevance for this project
OBC	STM32 Nucleo / STM32H7-class MCU board	Real-time control, timers, ADC/SPI/I2C/UART, strong tooling and RTOS support.
RTOS	FreeRTOS or Zephyr	Supports deterministic tasks, health supervision and telemetry scheduling.
EPS monitor	TI INA226 / INA226-Q1	16-bit current/voltage/power monitor, I2C/SMBus, programmable averaging and alert support. TI lists 0-36 V common-mode sensing and -40 to 125 C operation for the active INA226. [cite]turn913950search2[
Solar charge	TI BQ24650-class controller	Provides solar input and MPPT support; supports 1-6 series cells and LiFePO4 as a catalog chemistry option. [cite]turn913950search5[
Battery	Protected LiFePO4 pack sized for testbench	Lower-energy laboratory demonstration with explicit protection/BMS and safe charging procedure.
Temperature	Precision digital sensors + thermistors	Multiple measurement points for battery, OBC, panel and ambient thermal behavior.
IMU	ADI ADIS16470 or Bosch BMI088	ADIS16470 offers factory calibration and low bias stability; BMI088 offers 16-bit accel/gyro and vibration robustness for demanding motion applications. [cite]turn913950search0[turn913950search1[
Magnetometer	Industrial 3-axis digital magnetometer	Supports magnetic-field vector monitoring and sensor plausibility checks.
Radio	Sub-GHz LoRa-class module or academic packet-radio transceiver	Provides a practical laboratory link. Protocol layer can be designed independently.
Ground station	Linux/Windows PC + USB radio/serial interface	Runs decoder, database and GUI.
Structure	1U aluminum/CNC or well-designed printed frame	Provides mechanical organization and visual similarity to a CubeSat. NASA notes both COTS and custom-machined/additively manufactured smallsat structures are common. [cite]turn453057search10[

Important design note: an INA226 is not an EPS by itself, and a solar charger is not a complete battery-management system. The protection, switching, current limits, safe charging, fusing, connector strategy and fault isolation must be designed as a system. Likewise, an industrial IMU can improve

measurement quality but does not make the entire spacecraft ADCS flight qualified.

8. Subsystem Design Details

8.1 Electrical Power Subsystem (EPS)

The EPS should be designed around measured power flow. Place a high-side current/voltage monitor on the battery path and, where useful, on major regulated rails. The INA226 is a strong laboratory choice because it can report current, bus voltage and calculated power through I2C, with configurable averaging and alert capability. TI currently lists the INA226 as active and specifies a 16-bit monitor with 0-36 V common-mode sensing and -40 to 125 C operation. [cite]turn913950search2[

Recommended EPS telemetry: battery voltage, battery current, instantaneous battery power, solar input voltage/current, 3.3 V rail voltage/current, 5 V rail voltage/current, charger state, battery temperature, and cumulative energy-in/energy-out counters. A power budget should be recorded in the report so that each load has an expected current range and the measured values can be compared against it.

8.2 Thermal Control and Monitoring (TCS)

A laboratory CubeSat does not reproduce orbital thermal-vacuum conditions, but it can still investigate thermal telemetry, sensor calibration, heat paths, and fault detection. Place sensors near the battery, MCU, regulator, radio and one or more structure/panel locations. Use a controlled heat source and airflow changes to create repeatable disturbances.

8.3 On-Board Computer (OBC)

An STM32-class microcontroller provides a better academic platform than a Wi-Fi-first hobby board when the primary objective is spacecraft-like telemetry and deterministic control. The OBC software should be decomposed into tasks for sensor acquisition, telemetry production, health evaluation, communications, command processing, logging and watchdog supervision. FreeRTOS or Zephyr can be used to make task timing measurable.

8.4 Attitude Determination and Control (ADCS) Sensing

The ADIS16470 is particularly attractive when the project needs premium sensor data and a meaningful calibration/drift study; Analog Devices specifies a triaxial digital gyroscope, triaxial accelerometer, factory calibration, self-test functions, SPI and an operating range of -25 C to +85 C. The BMI088 is a lower-cost alternative designed for demanding drone/robotics applications, with 16-bit acceleration and gyro outputs, programmable rates and vibration robustness. Neither device alone forms an ADCS: attitude estimation and, if desired, control actuators must be treated as separate functions.

[cite]turn913950search0[turn913950search1[

8.5 Communications (COMMS)

For a college prototype, begin with a reliable wired UART/USB path and then add an RF link. This lets the team validate the telemetry protocol before radio effects are introduced. The packet layer should contain at least a synchronization pattern or framing strategy, version, spacecraft ID, packet type, sequence number, source timestamp, payload length, payload, and CRC. For the research paper, the packet design can be mapped conceptually to CCSDS Space Packet and space data-link concepts without claiming CCSDS compliance unless every applicable requirement has actually been implemented and tested. CCSDS identifies 133.0-B-2 as the current Space Packet Protocol Blue Book in the active publication set. [cite]turn453057search1[

9. Telemetry Packet Design

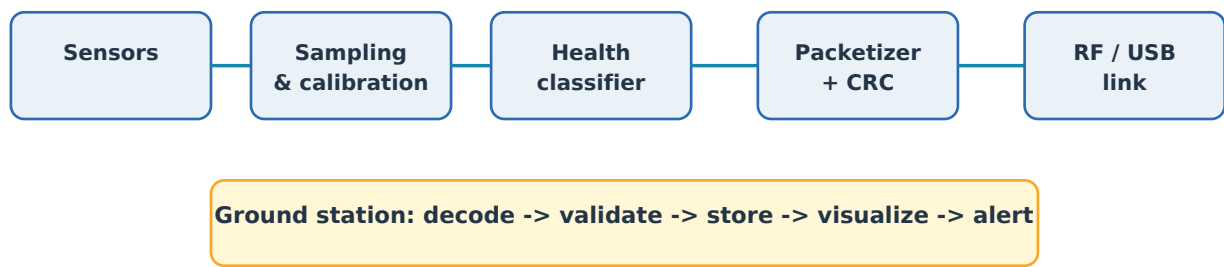


Figure 2. Telemetry pipeline: sensing, calibration, health classification, packetization and link transport, followed by ground-station processing.

Field	Example	Purpose
Sync / framing	0x55AA or link-specific delimiter	Find packet boundaries on a byte stream.
Version	1 byte	Allows future packet format revisions.
Spacecraft ID	2 bytes	Identifies the source node.
Packet type	1 byte	Beacon, housekeeping, event, payload status, command response.
Sequence	2 bytes	Detects drops, duplicates and reordering.
Timestamp	4-8 bytes	Time correlation between source and ground.
Payload length	2 bytes	Supports variable payloads safely.
Payload	Binary telemetry	Engineering measurements and flags.
CRC	2-4 bytes	Detects transmission corruption.

10. Ground Station GUI Design

The GUI should behave like an engineering console rather than a decorative dashboard. The primary screen should answer five questions immediately: Is the spacecraft connected? Is the spacecraft safe? What subsystem is abnormal? What changed recently? Can the operator trust the data?

Feature	Recommended content
Mission status bar	Connection state, last packet time, overall health, current mode, packet-loss indicator.
EPS card	Battery voltage/current/power, SoC estimate, solar input, bus rails, temperature.
Thermal card	Battery, OBC, radio and structure temperatures with alarm coloring.
ADCS card	Gyro/accel/mag values, attitude estimate (optional), sensor validity.
COMMS card	RSSI, SNR/link quality where available, CRC failures, sequence gaps, queue depth.
OBC card	CPU load, RAM, storage, task heartbeat, uptime, resets, watchdogs.
Event pane	Timestamped warnings, critical events, acknowledgements and command responses.
Trends tab	Selectable time-series plots with zoom, cursor and export.
Command panel	Safe, explicit telecommands with validation and confirmation.
Diagnostics tab	Raw frames, decoded fields, packet statistics, sensor validity and calibration details.

Recommended software stack: Embedded: C/C++ + STM32 HAL + FreeRTOS or Zephyr; Ground station: Python + PySide6/PyQt + PyQtGraph + SQLite. A web stack is also possible, but a desktop Python application reduces deployment complexity and makes direct serial/radio integration straightforward. The ground station should have a test mode that replays logged telemetry so the GUI can be validated

without live hardware.

11. Software Architecture and Data Model

Module	Responsibility
Sensor drivers	Acquire raw data and expose timestamps/errors.
Calibration	Convert raw counts to engineering units; store coefficients/version.
Health engine	Apply range, rate, persistence, hysteresis and sensor-validity rules.
Telemetry manager	Schedule packets, serialize payloads and maintain sequence numbers.
Protocol layer	Frame packets and validate CRC.
Command manager	Validate and execute safe telecommands; produce acknowledgements.
Data logger	Persist raw and decoded telemetry plus event records.
GUI model	Transform decoded telemetry into view-ready state.
Replay/test engine	Replay known-good and fault-injection datasets for regression testing.

A useful telemetry record in the database can contain: timestamp, packet sequence, parameter ID, value, engineering unit, health state, source packet, validity flag and optional calibration version. Event records should capture event time, subsystem, severity, code, message, acknowledgement state and associated telemetry snapshot.

12. Fault Injection and HIL Validation

HIL is the core feature that elevates the project from a dashboard into an engineering study. The testbench should intentionally generate faults while recording both the actual injected condition and the response of the embedded and ground systems.

Fault	Injection method	Expected result	Primary metric
Battery undervoltage	Programmable load or controlled supply reduction	Warning/critical transition; event generated	Detection latency
Overtemperature	Controlled heater near sensor	Warning, then critical if sustained	Threshold accuracy / false alarms
Sensor disconnect	Unplug or digitally suppress sensor	INVALID state; no stale data accepted	Invalid-data detection rate
Packet loss	Drop selected packets in transport layer	Sequence gaps detected	Packet-loss estimation accuracy
CRC corruption	Flip payload bits	Packet rejected	CRC detection rate
OBC task stall	Suspend heartbeat update	Watchdog/event indicates fault	Fault detection latency
High CPU load	Synthetic compute stress	CPU warning / scheduling margin drops	Jitter and missed deadlines
Radio interference	Introduce attenuation/noise in lab link	RSSI/link quality degrades; retries/loss observed	Link performance
Storage pressure	Fill log volume	Storage warning and controlled behavior	Remaining-capacity prediction

At least 30 repeated trials per major fault class are recommended when practical. Report mean, median, standard deviation and worst-case detection latency. For binary event detection, include false-positive and false-negative counts. This makes the work statistically defensible rather than anecdotal.

13. Experimental Program and Research Questions

ID	Research Question
RQ1	How does telemetry packet loss affect the correctness and latency of ground-station health decisions?
RQ2	How much does hysteresis/persistence reduce alarm chatter without materially delaying fault detection?
RQ3	What is the end-to-end latency from a physical sensor disturbance to a GUI alarm?
RQ4	How accurately can the system detect sensor faults using range, rate-of-change and heartbeat checks?
RQ5	How does OBC CPU load and telemetry frequency affect missed deadlines and power consumption?
RQ6	Can a composite subsystem health score improve operator prioritization without masking critical single-parameter faults?

A strong paper should select one primary research question and treat the remaining questions as supporting analyses. A project is more publishable when it makes one testable claim and provides repeatable measurements, rather than listing many features without an experimental hypothesis.

14. Example Limit-Management Strategy

Exact electrical and thermal limits must be based on the final selected hardware, battery chemistry, regulator design and environmental test conditions. Therefore the report should not present generic values such as “40% SoC = Warning” as universal spacecraft limits. Instead, define limits from datasheets, battery characterization and mission-level safety margins.

Parameter	Warning/Critical Threshold	Source/Justification
Battery voltage	Nominal band + Warning + Critical based on selected LiFePO4 pack and BMS limits	Battery/BMS datasheet + characterization
Battery temperature	Warning/critical margins inside charge/discharge limits	Battery + charger datasheets
3.3/5 V rails	Tight regulation window around nominal rail	Regulator/OBC datasheets
CPU load	Warning at sustained high utilization; Critical when real-time deadlines are threatened	Measured task timing
Sensor temperature	Limits below component operating maximum with project margin	Sensor/OBC/radio datasheets
Packet loss	Warning based on mission link budget and application tolerance	Measured link statistics
Storage	Warning before reserve threshold; Critical before logging failure	Filesystem/logger tests

15. Verification and Acceptance Matrix

Requirement	Verification Method	Acceptance Criteria
Telemetry sampling	Bench measurement + log review	All required parameters update at specified rate within tolerance.
Calibration	Compare against traceable reference instruments	Error within defined project tolerance.
Packet integrity	CRC fault injection	Injected corruption detected.
Sequence tracking	Drop/duplicate packet test	All missing or duplicate sequences identified.
Health transitions	Controlled threshold crossing	Correct state reached within defined persistence time.
Alarm recovery	Return parameter inside recovery band	State returns without excessive chatter.
GUI latency	Timestamped end-to-end trial	Latency measured and reported statistically.
Data persistence	Long-run logging test	No silent data loss; recovery behavior documented.

Requirement	Verification method	Pass criterion
Telecommand safety	Invalid-input and duplicate-command tests	Unsafe or malformed commands rejected safely.
Power budget	Measure subsystem currents	Measured profile consistent with design budget.

16. Engineering Risks and Mitigations

Risk	Impact	Mitigation
Battery mishandling	Safety + hardware damage	Use protected pack/BMS, correct charger profile, fuse, enclosure and supervised testing.
False alarms	Poor operator trust	Use persistence/hysteresis; validate on real datasets.
Stale telemetry	Incorrect health decisions	Timestamps, freshness timeout and INVALID state.
Clock drift	Incorrect trend correlation	Periodic synchronization or monotonic source time.
RF complexity too early	Project schedule risk	Validate protocol over wired link before RF.
Overclaiming “space grade”	Research credibility risk	Clearly distinguish COTS prototype from qualification/flight hardware.
GUI-only intelligence	Loss of autonomy / inconsistent results	Keep health rules in a testable embedded data model and reproduce them on ground.

17. Relevant Standards and References

The following sources should be cited directly in the final academic manuscript. They establish the CubeSat context, smallsat mission-success considerations, telemetry/packet standards and component specifications used in the proposed design.

NASA, “What are SmallSats and CubeSats?” Updated 5 February 2026. Defines the CubeSat class, nominal 1U dimensions and common larger form factors. [\[cite\]turn453057search7](#)

NASA GSFC, “Mission Success Handbook for CubeSat Missions,” GSFC-HDBK-8007, Baseline, 16 December 2019. Provides mission-success guidance under the size, cost and schedule constraints of CubeSat missions. [\[cite\]turn453057search4](#)

NASA SmallSat Institute, “Structures, Materials, and Mechanisms - CubeSat Standard.” Current smallsat state-of-the-art guidance on COTS/custom structures and standard structure dimensions. [\[cite\]turn453057search10](#)

CCSDS, “Space Packet Protocol,” CCSDS 133.0-B-2, Blue Book, Issue 2, June 2020. Active Recommended Standard for CCSDS Space Packets. [\[cite\]turn453057search1](#)

Texas Instruments, “INA226 36V, 16-Bit, Ultra-Precise I2C Output Current, Voltage, and Power Monitor With Alert,” active product documentation. [\[cite\]turn913950search2](#)

Texas Instruments, “BQ24650 Stand-Alone Synchronous Buck Battery Charge Controller for Solar Power With Maximum Power Point Tracking,” product documentation. [\[cite\]turn913950search3](#)[\[turn913950search5\]](#)

Analog Devices, “ADIS16470 Wide Dynamic Range Miniature MEMS IMU,” product page and datasheet. [\[cite\]turn913950search0](#)[\[turn913950search84\]](#)

Bosch Sensortec, “BMI088 High-performance Inertial Measurement Unit,” product documentation. [\[cite\]turn913950search1](#)[\[turn913950search85\]](#)

18. Suggested Research-Paper Structure

Section	Recommended content
1. Introduction	CubeSat context; health monitoring problem; motivation; research gap.
2. Related Work	Small-satellite telemetry, health management, fault detection, HIL and ground systems.
3. System Architecture	Hardware, data flow, software layers and interfaces.
4. Telemetry and Health Model	Parameters, sampling rates, packet design, limits, persistence, hysteresis.

Section	Recommended content
5. Prototype Implementation	BOM, PCB/wiring, enclosure, embedded software and GUI.
6. Experimental Methodology	Test setup, fault injection, datasets, repetitions and metrics.
7. Results	Detection latency, false alarms, packet integrity, GUI latency, power and CPU load.
8. Discussion	Interpretation, limitations, comparison and trade-offs.
9. Conclusion	Research findings and next steps toward a more flight-representative system.

19. Suggested Initial Bill of Materials (Prototype-Level)

Item	Quantity	Note
STM32 Nucleo / evaluation board	1	OBC / development platform
INA226 breakout or custom sensing board	2-4	Battery and rail monitoring
LiFePO4 protected battery pack + BMS	1	Energy storage for bench system
BQ24650-based charger stage or suitable development board	1	Solar charging / MPPT study
ADIS16470 evaluation module or BMI088 board	1	IMU / ADCS sensing
3-axis magnetometer board	1	Magnetic sensing
Digital temperature sensors / thermistors	4-8	Battery, OBC, radio, structure and ambient
Sub-GHz/LoRa-class radio modules	2	Air link / ground station
MicroSD or eMMC storage	1	Onboard event / telemetry log
USB-UART / debug interface	1	Bench bring-up and fallback telemetry link
1U-class frame / panels	1	Mechanical integration
Bench PSU + electronic load	1 each	Controlled power and fault injection
Logic analyzer / oscilloscope	1 each	Timing, buses and debugging

20. Conclusion

The proposed CubeSat Health Monitoring System is a credible high-level college engineering project because it integrates embedded systems, power electronics, sensing, communications, software engineering, data management and human-machine interaction into a single measurable testbed. The central research opportunity is not simply to build a dashboard; it is to demonstrate that the complete chain from physical disturbance to trustworthy health decision can be characterized, validated and improved.

The recommended path is to begin with a wired telemetry prototype, establish the data model and health engine, add calibrated EPS/TCS/ADCS sensors, implement packet integrity and event logging, then introduce the RF link and full GUI. The final phase should be fault-injection testing with repeated trials and statistical analysis. This sequencing reduces integration risk while preserving a strong research narrative.

The system should be described throughout the final paper as a CubeSat-inspired engineering demonstrator or hardware-in-the-loop small-spacecraft testbed, unless and until formal qualification evidence exists for flight use. That distinction increases, rather than decreases, the credibility of the project because it makes the experimental scope explicit.

Appendix A - Example Telemetry Dictionary

ID	Param	Unit	Rate	Health Use
0x01	Battery voltage	V	10 Hz	UV/OV + trend
0x02	Battery current	A	10 Hz	Charge/discharge
0x03	Battery temperature	C	2 Hz	Thermal safety
0x04	Solar input voltage	V	2 Hz	Generation
0x05	Solar input current	A	2 Hz	Generation
0x06	3.3 V rail	V	10 Hz	Regulation
0x07	5 V rail	V	10 Hz	Regulation
0x08	CPU load	%	1 Hz	Resource margin
0x09	Free memory	bytes	1 Hz	Resource margin
0x0A	Storage used	%	1 Hz	Capacity
0x0B	Reset count	count	event	Fault history
0x0C	Watchdog resets	count	event	Fault history
0x10	Gyro X/Y/Z	deg/s	100 Hz local / 10 Hz TM	ADCS health
0x11	Accel X/Y/Z	g	100 Hz local / 10 Hz TM	ADCS health
0x12	Mag X/Y/Z	uT	20 Hz local / 5 Hz TM	ADCS health
0x20	RSSI	dBm	1 Hz	Link health
0x21	CRC failures	count	1 Hz	Link integrity
0x22	Packet gaps	count	1 Hz	Link integrity
0x30	Payload status	enum	1 Hz	Instrument health

The rates above are example engineering values rather than fixed requirements. Local sensor sampling can be substantially faster than downlinked telemetry; the onboard system may retain higher-rate data for selected windows while transmitting lower-rate housekeeping summaries.

Appendix B - Deliverables Checklist

- ☐ System architecture diagram
- ☐ Electrical schematic and wiring harness diagram
- ☐ Embedded firmware repository and build instructions
- ☐ Telemetry dictionary and packet specification
- ☐ Health-state rules and configurable threshold table
- ☐ Ground-station GUI and database schema
- ☐ Calibration procedure and test records
- ☐ Fault-injection scripts/test fixtures
- ☐ Experimental dataset and statistical analysis
- ☐ Final engineering report / research paper